

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2021

ESE 3203

(Petroleum and Natural Gas Processing)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Is it better to burn natural gas or release them into atmosphere without burning? Show your reasons. 05
- 1(b). What will be your recommendations if there is any unconventional reservoir found in Bangladesh? Describe from an engineering perspective. 06
- 1(c). Briefly describe the theories regarding the origin of natural gas. 08
- 1(d). You have found a condensate gas reservoir. What will be your concerns regarding production of this gas? Also, how can you solve these issues as an energy engineer? Discuss with proper illustration. 16
- 2(a). A natural gas has the following composition: $C_1=80\%$, $C_2=3\%$, $C_3=2\%$, $n-C_4=2\%$, $i-C_4=1\%$, $n-C_5=1\%$, $i-C_5=3\%$, $n-C_6=0.05\%$, $n-C_{7+}=1\%$, $N_2=0.05\%$, $CO_2=2\%$ and $H_2S=4\%$. Assume the C_{7+} fraction to be equivalent to nC_{10} . The temperature and pressure of the gas are $180^\circ F$ and 1900 psia respectively. Find out its compressibility factor and viscosity. Use necessary chart and table and there is no need to show the pressure correction for the viscosity. 35
- 3(a). What is vortex? Why is it undesirable in gas-liquid separation? What can you do to avoid it? 10
- 3(b). In a natural gas plant there is a horizontal separator. Due to the reduction of production of natural gas, the authority plans to replace the separator so the operating cost can be minimised. Which separator will you prefer for the mentioned circumstances? Draw the schematic diagram of your selected separator. What are its advantages and disadvantages over the present separator? 15
- 3(c). Briefly describe “iron sponge unit” for the removal of H_2S with a schematic diagram. 10
- 4(a). What steps should you take as an engineer to avoid the formation of hydrate in pipeline? Describe with the help of equilibrium diagram of gas-water-hydrate system. 15
- 4(b). You have three types of absorbents which are alumina, silica gel and molecular sieve. Which one will you prefer for the following criteria during dehydration of natural gas? Criteria are (i). must handle sour gas, (ii). Should not be expensive, (iii). Should be able to dehydrate gas to 12 ppm. Show your reasons through a comparative analysis among the three absorbents. 10
- 4(c). For natural gas sweetening process you have to choose between iron sponge unit and solid bed sweetening process. The raw gas contains 3% CO_2 and 0.02% H_2S . Your budget is low for operating the sweetening process. Which process will you select? Enumerate with critical analysis between the two processes. 10

SECTION – B

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|-------|--|----|
| 5(a). | Why petroleum is called “fossil fuel”? Briefly discuss about the composition of petroleum with appropriate figures of commonly found hydrocarbon compounds in crude oil. | 15 |
| 5(b). | What is meant by “heat capacity”? Why dehydration is needed in crude oil refining process? | 08 |
| 5(c). | Explicate the adverse effects of sulphur compounds in crude oil refineries. Mention the test methods used for detecting sulphur compounds. Enumerate the types of stabilizers for crude oil stabilization. | 12 |
| | | |
| 6(a). | Briefly explain crude oil distillation process. | 15 |
| 6(b). | Mention the basic functions and types of catalytic cracking process. What is “spent catalyst”? | 08 |
| 6(c). | Write down the catalytic reforming reactions. | 12 |
| | | |
| 7(a). | Enunciate the construction of LNG carrier with appropriate figure. | 10 |
| 7(b). | Discuss the commercial and industrial uses of LPG. | 10 |
| 7(c). | What are the potential hazards and their safety legislation associated with LPG? | 15 |
| | | |
| 8(a). | Clarify the necessity of using lube oil in vehicles. | 08 |
| 8(b). | Tabulate major differences among LPG, LNG and CNG. | 12 |
| 8(c). | Describe the function of an energy engineer in a petroleum refinery. | 15 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2021

ESE 3221

(Energy Storage Systems)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|---|----|
| 1(a). Define energy density, power density and specific power. | 06 |
| 1(b). What is Ragone chart? Draw a Ragone chart for various energy storage systems. | 08 |
| 1(c). Explain the working principle of a pumped hydro energy storage system. List the various losses in this system. | 15 |
| 1(d). Classify the energy storage technologies. | 06 |
| 2(a). What is Compressed Air Energy Storage (CAES) technology? What are the main disadvantages of CAES? | 10 |
| 2(b). What are the applications of flywheel energy storage systems? | 06 |
| 2(c). What is thermal stratification? What are its effects on sensible thermal energy storage systems? | 08 |
| 2(d). How thermal energy can be stored in solid storage medium? Explain with schematic diagram. | 11 |
| 3(a). Explain the working principle of Superconducting Magnetic Energy Storage system with schematic diagram. | 13 |
| 3(b). What are the differences between battery and capacitor? Explain the working mechanism of electrochemical double layer capacitor with necessary diagram. | 15 |
| 3(c). What are the factors affecting the capacitance of a capacitor? | 07 |
| 4(a). What is phase change material? Explain its various properties to use it in a LHTS. | 08 |
| 4(b). Explain the differences between capacitor and super capacitor. | 08 |
| 4(c). What is hybrid power plant? Explain with schematic diagram. | 10 |
| 4(d). Explain the three types of UPS technologies. | 09 |

SECTION – B

5(a).	Explain the classification of battery? What are the main differences between primary and secondary cells?	09
5(b).	Explain the charging and discharging of lead-acid battery with chemical reaction.	08
5(c).	What is Solid Electrolyte Interphase (SEI) layer? Explain the impact of SEI layer on anode and cathode materials in Li-ion battery.	10
5(d).	“Cylindrical cell is more durable than pouch cell”-explain the statement.	08
6(a).	Explain the working principle of vanadium redox flow battery with thematic diagram and mention the current challenges associated with this battery.	15
6(b).	Why Pb-acid battery is used at starting of engine? Briefly mention the advantages and disadvantages of Pb-acid battery.	10
6(c).	What are the batteries with aqueous electrolytes? Describe the basic working principles of Ni-MH battery.	10
7(a).	What are the differences between fuel cell and battery? Classify the fuel cell based on temperature range.	07
7(b).	Describe the working principle of a PEM fuel cell with schematic diagram.	12
7(c).	What types of conventional fuels are used for H ₂ generation? Explain with reactions.	10
7(d).	What are the commercial fuel cells? Give some examples.	06
8(a).	What is a fuel cell stack? Illustrate with a thematic diagram connecting multiple fuel cells in a series to obtain practical voltage level.	10
8(b).	How would you differentiate between acid fuel cell and alkaline fuel cell with respect to electrodes, electrolytes, fuel input and output?	10
8(c).	Why metal hydrides is more attractive compared to compressed H ₂ gas storage technology? Explain.	08
8(d).	What is power fade and capacity fade in energy storage system?	07

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2021

ESE 3207

(Solar Photovoltaic Systems)

Time: 3 Hours

Full Marks: 210

N.B. i) Answer any three questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|---|----|
| 1(a). | What is the purpose of doping in a semiconductor? Explain different types of doping with Band diagram leveling Fermi level energy. | 10 |
| 1(b). | Briefly explain direct and indirect Band gap semiconductor. Describe the SRH and auger recombination process mechanism with their band diagram. | 15 |
| 1(c). | The total absorption in a $d = 300[\mu m]$ thick c-si wafer for light with a wave length of $500[nm]$. Incident photons are reflected from the surface is 0.38 and absorption coefficient $\alpha = 1.11 \times 10^4 [cm^{-1}]$. The incident irradiance is $1000[Wm^{-2}]$. Calculate the photocurrent density. | 10 |
| 2(a). | Mention the names of some iii-v materials used in thin-film solar cell production. Also discuss about the design of thin-film silicon solar cells. | 12 |
| 2(b). | What is meant by Lithography? Write down the steps of Lithography with proper diagram. | 11 |
| 2(c). | Briefly explain the production of C-Si wafer from quartzite. | 12 |
| 3(a). | “Performance of solar cell depends on irradiance” Justify the statement. | 08 |
| 3(b). | A crystalline silicon wafer is doped with 10^{16} acceptor atoms per cubic centimetre. A, 1 micrometer thick emitter layer is formed at the surface of the wafer with a uniform concentration of 10^{18} donors per cubic centimetre. Assume a step p-n junction and that all doping atoms are ionized. The intrinsic carrier concentration in silicon at 300 K is $1.5 \times 10^{10} cm^{-3}$. Calculate the electron and hole concentrations in the p- and n-type quasi-neutral regions at thermal equilibrium. | 12 |
| 3(c). | What is lattice mismatch? Justify the use of Ge, GaAs and GaInP to form triple junction cells. | 10 |
| 3(d). | What are the criteria for materials to be used in solar cell. | 05 |
| 4(a). | Draw an equivalent circuit of an Ideal solar cell and a solar cell with series resistance (R_s) and shunt resistance (R_p), Mentioning their modeled equation. | 09 |
| 4(b). | Briefly explain the efficiency losses of a solar cell. Also define internal quantum efficiency and external quantum efficiency. | 10 |
| 4(c). | What are the approaches to increase efficiency of solar cell? Explain how multi-junction solar cell increases efficiency compared to single junction solar cell. | 10 |
| 4(d). | Write short note on generation of solar cell. | 06 |

SECTION – B

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|-------|---|----|
| 5(a). | Write short notes on: i). PV cell, ii) PV module, iii) PV array | 06 |
| 5(b). | Describe the PV system's shedding effect. Give a justifiable explanation of how the PV system performed when there was shedding. | 10 |
| 5(c). | Explain the working principle of photovoltaic panel in series and parallel connection. Why series and parallel connections are necessary in PV systems? | 08 |
| 5(d). | A solar panel with dimensions $61\text{cm} \times 46\text{cm}$ is illuminated at standard test conditions. The cell has the following external parameters: $V_{OC} = 21.5\text{V}$, $I_{SC} = 3.21\text{A}$, $V_{MPP} = 20\text{V}$, $I_{MPP} = 2.5\text{A}$. Calculate the fill factor and efficiency of this solar panel. | 11 |
| 6(a). | What is carbon footprint of making solar panels? Explain the environmental impact of PV systems. | 08 |
| 6(b). | What is considered the most important feature in a charge controller? How do solar MPPT charge controllers work? Explain in details. | 12 |
| 6(c). | A house has the 3(Three) 60w lights, 2(Two) 90w fans, 1(one) 60w radio AC loads with rated at 230V. All the loads are powered simultaneously. A 12V battery is available. Choose the appropriate inverter? Assume that inverter efficiency is 90%. | 08 |
| 6(d). | How does heat affect solar panel efficiency? Describe. | 07 |
| 7(a). | What are the properties and characteristics of Perovskite solar cells? Write down the classifications of perovskite solar cells. Explain the basic structure of Perovskite solar cells. | 14 |
| 7(b). | What is Net Metering? Write down the eligibility conditions for a customer to be considered for Net Metering connection Bangladesh. | 08 |
| 7(c). | How to design and sizing of solar photovoltaic systems and installation of solar PV system. | 13 |
| 8(a). | How does a grid connected PV system work? Write the conditions for grid interfacing of On-grid PV system. | 08 |
| 8(b). | What is total harmonic distortion in solar inverters? How can reduce total harmonic distortion in inverters? Discuss the different types of solar inverters based on output waveform. | 09 |
| 8(c). | What are the basic parameters of a DC to DC converters? Write the basic working principles of DC-DC conversion of solar PV system. | 08 |
| 8(d). | How does the concept of levelized cost of electricity work? Describe. | 10 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2021

ESE 3217

(Instrumentation and Control)

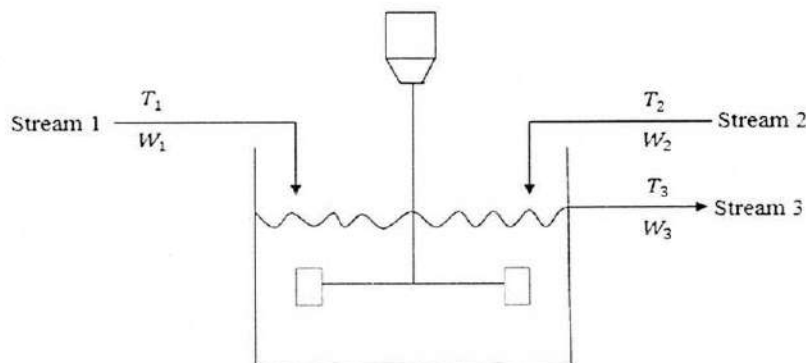
Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
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iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is measurement system? Discuss a measurement system with necessary diagram. 10
1(b). Discuss the four different ways that a dynamic model is used in process control engineering. 10
1(c). A perfectly stirred, constant volume tank has two input streams as shown in figure, both consisting of the same liquid. The temperature and flow rate of each of the streams can vary with time. 15



- i. Derive a dynamic model that will describe transient operation of the system.
ii. Make a degree of freedom analysis assuming that both streams 1 and 2 comes from upstream units (i.e., their flow rates and temperatures are known functions of time).
- 2(a). For the equation, $\ddot{y} + 5\dot{y} + 6y = 7$, where $\dot{y}(0) = 0, y(0) = 1$. Use the partial fraction method to find $y \equiv y(t)$. 15
2(b). Find the transfer function for the following ODE: $\ddot{X} + 5\dot{X} + 4X = u(t)$ where $\dot{X}(0)=0, X(0) = 0$ 10
2(c). Linearize the following ODE: $\frac{dX}{dt} = -X^2 + \sqrt{u}$ at $X_0 = 2, u_0 = 16$ 10
- 3(a). What is PID controller? Describe the PID controller operation with corresponding equations and necessary graphs. 12
3(b). Define poles and zeros by considering an nth order linear ODE with m order forcing term. 05
3(c). An electrically heated, stirred tank model is given in terms of deviation variable. Where $T'_i = 0$; 18

$$\frac{dT'_i}{dt} = 0.1(T'_i - T') + 0.1(T'_e - T')$$

$$\frac{dT'_e}{dt} = 0.05Q' - (T'_e - T')$$

- i. Find the state space model of the system.
ii. Also find the eigen values of the system and comment wheather the system stable or unstable.
- 4(a). Why calibration of measuring instruments is important? 05
4(b). How to calculate the dynamic error of a measuring system? 06
4(c). Briefly explain the working principle of a control valve. 11
4(d). What should be the most important consideration in selecting of a measurement device? Discuss it. 07
4(e). Show the general procedure for developing transfer function models. 06

SECTION – B

- | | | |
|-------|---|----|
| 5(a). | Distinguish between a digital signal and an analog signal. Describe the essential steps of the communication process. | 10 |
| 5(b). | What are the most common type of cable used for control communication? How does a fiber optic cable work? Explain. | 09 |
| 5(c). | Where do instrument engineer work? What are the skills required for instrumentation engineer. | 06 |
| 5(d). | Write short note on microcontroller. Draw the pin diagram of a 8051 microcontroller and write the function of each pin. | 10 |
| | | |
| 6(a). | What types of Arduino are available and how do they differ? How to make an LED light up and blink using Arduino and write Arduino code. | 12 |
| 6(b). | Why are SCADA systems implemented in the first place? How does SCADA system work? Describe. | 12 |
| 6(c). | What is industrial automation and what are its components? Why is the cyber security of industrial automation and control systems important? | 11 |
| | | |
| 7(a). | Make a distinction between sourcing and sinking in PLC. Draw the basic internal architecture of a PLC and write the functions of different types of bus in PLC. | 12 |
| 7(b). | How does a PLC communicate with field devices such as sensors, actuators, DCS, SCADA or other PLCs? What are some of the common communication protocols used by PLC systems? | 07 |
| 7(c). | Create a PLC program for automatic water tank level control in which one light is illuminated when pump is running and another light is illuminated when the pump is turned off. | 10 |
| 7(d). | What is the function of latching in PLC and battery-backed relays? | 06 |
| | | |
| 8(a). | How to use JUMP and CALL instruction in PLC ladder logic? Describe it with an example. | 08 |
| 8(b). | Why timers are essential in a PLC system? Explain the different type of timers. | 08 |
| 8(c). | What is master control relay of a PLC? Write a PLC program use three sensors and control fire alarm systems. | 09 |
| 8(d). | Design the ladder program to implement automation of parking garage system in an apartment where they can park only 10 cars. When 10 cars park, a signal will turn ON signaling that a garage is full and notifying other devices not to enter because there is no space available. | 10 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 2nd Term Examination, 2021

ESE 3211

(Coal Power Generation)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- | | | |
|-------|---|----|
| 1(a). | What are the different types of coal? Write down the advantages and disadvantages of bituminous coal. | 08 |
| 1(b). | Discuss about the contribution of coal to global and Bangladesh electricity generation. | 10 |
| 1(c). | Define: i) coal geology, ii) coal reclamation. | 04 |
| 1(d). | Write down the significance of proximate and ultimate analysis of coal. A subbituminous coal has the elemental composition : 75% C, 4% H, 19% O, 1.5 % N and 0.5% S. Calculate its atomic hydrogen-to-carbon ratio. | 13 |
| | | |
| 2(a). | Why coal and ash handling are necessary? Describe the working principle of positive – pressure dilute- phase system with necessary figure. | 12 |
| 2(b). | What is coal preparation and coal storage ? Write down the steps carried out in coal preparation. | 06 |
| 2(c). | What are bottom ash and fly ash? Write down the useful applications of bottom ash and fly ash. | 07 |
| 2(d). | What are the different coal mining methods? With the help of necessary sketch, discuss the drift mining of coal. | 10 |
| | | |
| 3(a). | Why size reduction of raw coal is necessary? With the help of schematic diagram, describe the working principle of roll crusher used for size reduction of coal. | 10 |
| 3(b). | Define dust collection system. Describe the working principle of settling chamber used for dust collection with the help of necessary figure. | 10 |
| 3(c). | What are the different parts of fabric filters? Describe the working of centrifugal dust collection system. | 10 |
| 3(d). | Describe the selection criterion of a dust collector? | 05 |
| | | |
| 4(a). | Write short notes on: i) coal devolatilization, ii) coal firing. | 06 |
| 4(b). | Write down the functions of stokers. What are the limitations of tangential burner. | 06 |
| 4(c). | What are the differences between fixed bed and fluidized bed coal combustion? With the help of schematic diagram, describe the working principle of Pressurized Fluidized Bed combustion. | 13 |
| 4(d). | What is meant by pulverized coal combustion? Write down its merits and demerits. | 10 |

SECTION – B

- 5(a). Describe the necessities of clean coal technology. 07
- 5(b). How is CO_2 captured before and during combustion of coal? Explain. 08
- 5(c). What are the methods used for CO_2 capture from flue gas? Explain a chemical absorption system of CO_2 capture. 10
- 5(d). Write down the importance of CO_2 sequestration. Is it possible to reduce CO_2 emission by increasing the efficiency of thermal cycles? Briefly describe. 10
- 6(a). Why the emission of Mercury should be controlled? Describe the sorbent injection process to control the emission of Mercury. 12
- 6(b). What are the technologies used to control the emission of NO_x . Write short note on the following terms: 08
- i) Subcritical thermal power plant.
 - ii) Supercritical thermal power plant.
 - iii) Ultra-supercritical thermal power plant.
- 6(c). What is boiler specification? 15
- In a steam power plant, following observations were made on boiler during one hour test: steam pressure = 18 bar, steam temperature = 250°C, Temperature of water entering the economizer = 20°C, Temperature of water leaving the economizer = 95°C, Dryness fraction of steam leaving the boiler = 0.95, Mass of steam generated = 4500 kg, Mass of fuel used = 500 kg, Calorific value of fuel = 29000 kJ/kg. Calculate: i) the equivalent evaporation rate per kg of fuel, ii) Overall efficiency of the plant, and iii) the percentage of the available heat utilized in the boiler, economizer and superheater respectively.
- 7(a). What is meant by boiler? Describe the necessities of using mountings and accessories in boiler. 04
- 7(b). With the help of necessary sketch, describe the working principle of an advanced pulverized coal-fired power plant. Why it is called Advanced coal-fired power plant? 12
- 7(c). What are steam turbine, impulse turbine and reaction turbine? Why nozzles are used in steam turbine? 08
- 7(d). The following particulars refer to a single row impulse turbine: 11
- Mean diameter of blade ring = 2.5 m, speed 3000 rpm, nozzle angle = 20°, ratio of blade velocity to steam velocity = 0.4, blade friction factor = 0.8, blade angle at exit = 3° less than that at inlet, steam flow = 36000 kg/h. Draw velocity diagram for moving blade and estimate i) power developed, ii) blade efficiency and iii) steam consumption in kg/kWh.
- 8(a). Why compounding of turbine is necessary? For a four stage steam turbine, with the help of h-s diagram, show the condition line. Also, from h-s diagram, mathematically define the reheat factor. 09
- 8(b). Write short notes on: i) Degree of reaction, ii) Steam turbine governing. 06
- 8(c). Discuss about the different losses in steam turbine. 10
- 8(d). Steam enters a 0.5 degree of reaction turbine at 1.4 MPa and 315 °C and is expanded to a pressure of 15 kPa. The turbine has a stage efficiency of 75% for each stage and the reheat factor is 1.04. The turbine has 20 successive stages and the total power output is 12 MW. Assuming that all stages develop equal work, calculate the steam flow rate. (You may use Mollier diagram) 10

-End-